

StreetRace Manager Integration Testing Report

Total integration test cases: 154

Passed: 154

Failed: 0

1. System Overview

StreetRace Manager is a command-line system made of multiple interacting modules. The integration task checks that data and business rules remain correct when the modules are used together.

2. Modules Implemented

- `registration.py`: registers crew members and validates names and roles.
- `crew_management.py`: updates role, skill level, and crew availability.
- `inventory.py`: manages cars, inventory items, and cash balance.
- `race_management.py`: creates races, assigns cars and drivers, and starts races.
- `results.py`: records results, updates cash, frees resources, and maintains rankings.
- `mission_planning.py`: creates missions, assigns available crew by role, and completes or fails missions.
- `garage.py`: repairs damaged cars using mechanics and records repair activity.
- `leaderboard.py`: builds leaderboard views and mission success metrics.
- `models.py` and `store.py`: shared data layer used across all modules.
- `main.py`: command-line interface for the integrated system.

Required modules are Registration, Crew Management, Inventory, Race Management, Results, and Mission Planning. The two additional modules are Garage and Leaderboard.

3. Call Graph Requirement

The hand-drawn call graph should show calls within and between modules, especially registration to crew management, race management to results to inventory, mission planning to crew management, and garage to inventory. The repository contains graph source material in `integration/diagrams/`, but the final submission still needs a hand-drawn image there.

4. Integration Test Design

The integration tests were written around realistic workflows and assignment business rules.

- Register a driver, add a car, create a race, assign both, start the race, record the result, and verify prize money updates inventory cash.
- Reject a mechanic when trying to assign a race driver.
- Reject an unregistered member assignment to a race.
- Reject damaged-car race assignment.
- Reject mission assignment when required roles are unavailable.
- Require a mechanic for damaged-car mission scenarios.
- Verify successful missions reserve and later release crew.
- Verify drivers become available again after race completion.
- Verify leaderboard ordering after multiple races.
- Verify repaired cars can later be used in races.
- Verify role change from strategist to driver enables race entry.
- Verify mission and race availability conflicts are enforced.

5. Verified Business Rules

- A crew member must be registered before system workflows can use that member.
- Only drivers can be entered in races.
- Unavailable drivers cannot be assigned to races.
- Damaged cars cannot enter races until repaired.
- Race completion updates both rankings and inventory cash balance.
- Mission assignment fails when required roles are unavailable.
- Crew reserved by races or missions are released correctly afterwards.

6. Final Integration Audit Result

The integration portion is in good technical shape. The required and extra modules are present, the tests pass, and the test suite covers the important cross-module rules described in the assignment. The remaining manual item is the hand-drawn call graph image.